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Industrial smart product-service systems solution design via hybrid concerns

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Abstract

Enabled by the third and fourth waves of Information Technology (IT) innovation, together with the sustainable concerns towards a circular economy, manufacturing industries are shifting rapidly towards a service-oriented paradigm in the smart, connected environment, i.e. smart product-service systems (Smart PSS). Owing to its unique characteristics, it reshapes the solution design (product-service design) with several hybrid concerns. This research, as an explorative study, underlines three general ones at the macro-level, i.e. design and redesign as an overall value generation process (hybrid design), human and machine intelligence integration (hybrid intelligence), and technical and business innovation (hybrid value). Accordingly, novel design methods, intelligent systems, and value co-creation models are envisioned at the micro-level to assure industrial Smart PSS solution design success. To make our visions concrete, two prospective application scenarios leveraging abovementioned hybrid concerns are given at last. It is hoped this work can contribute to the future directions of this interdisciplinary research area, and offer insightful guidance to industrial companies in their solution design process with profitable outcomes.

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Keywords: Smart product-service system; cloud-edge computing; mobile crowd sensing; value co-creation; open innovation

1. Introduction

Nowadays, industrial companies are overwhelmingly embracing the third-wave (i.e. Internet-of-Things (IoT)) and fourth-wave (e.g. Artificial Intelligence (AI), Cyber-Physical Systems (CPS)) of IT competition to obtain high revenue with more sustainable concerns [1]. Servitization and digitalization represent the main trend, where smart, connected products (SCPs) and their generated e-services are considered as a single solution bundle to achieve individual customer satisfaction [2]. Reversely, users are actively involved in the value creation process by contributing their own data or knowledge (e.g. crowd sensing) [3]. This IT-driven value co-creation business strategy is known as smart product-service systems (Smart PSS) [4].

In this context, unlike conventional design process starts from the very beginning of lifecycle, Smart PSS solution (product-service) design can be regarded as a value generation

process by considering both value creation at the early development stage and value recreation in the usage stage, as the *hybrid design* [5]. Meanwhile, SCPs own the powerful computation and communication capabilities with various built-in sensors that allow them to generate data and communicate to the Internet [6]. Hence, massive user-generated and product-sensed data can be leveraged, as *hybrid intelligence*, to extract/generate valuable solution design concepts. Moreover, as an IT-driven business strategy, both technical and business innovation, as the *hybrid value*, should be considered integrally [4]. Nevertheless, as an emerging topic, to the authors best knowledge, scarcely any work reports on the Smart PSS solution design, let alone to describe it from hybrid concern perspectives.

This research work, as the first attempt, underlines the abovementioned three general hybrid concerns in the Smart PSS solution design phase, and further proposes several novel methodologies accordingly to ensure solution design success

in the industrial environment. The rest of this paper is organized as follows: Section 2 outlines the fundamentals of Smart PSS to distinguish it from the others. Section 3 describes the three hybrid concerns in the solution design phase, and Section 4 proposes the prospective approaches to handle those concerns. Section 5 further validate those methodologies with two promising industrial application scenarios. The conclusion and future work are summarized in Section 6 at last.

2. Fundamentals of Smart PSS

Smart PSS, is defined as an “an IT-driven value co-creation business strategy consisting of various stakeholders as the players, intelligent systems as the infrastructure, smart, connected products as the media and tools, and their generated e-services as the key values delivered that continuously strives to meet individual customer needs in a sustainable manner” [4]. From the definition, one can find that its key elements are *stakeholders*, *SCPs* and *e-services*, and *intelligent systems*, while *value co-creation with sustainable concerns* is the business value proposition.

Stakeholders as the value co-creation participants, who can be mainly classified into three groups, i.e. users, service providers and manufacturers & vendors.

- *Users* are the people who demand or consume the value.
- *Service providers* refer to the companies/individuals who design, share or maintain the services offered.
- *Manufacturers & vendors* represent all the parties involved in the SCP manufacturing process.

SCPs and e-services are composed by three parts, i.e. physical components, smart and connectivity components, and SCP generated e-services [4]. The first two are the inherent elements of a SCP [7], where 1) *physical components* comprising the electrical and mechanical parts as usual product's; 2) *smart components* consist of the sensors, microprocessors, data storage, controls, software, and, typically, an embedded operating system and enhanced user interface; and 3) *connectivity components* represent the ports, antennae, and protocols enabling wired or wireless connections with the product [3]. Hence, they are capable of collecting, processing, produce and exchange information and react to the environment to some extent. At the same time, unlike the usual services without intelligence and connectedness (e.g. offline maintenance service), e-services mainly include two folders: 1) add-on services, which are independent with the product components (e.g. broadcasting APPs in the smartphone); and 3) digitalized services, which are the ones largely dependent with the product itself (e.g. digital twin-enabled machine tool) [1].

Intelligent system or smart environment [1] refers to a broad scope of advanced IT, which can assure high connectivity and smartness. For instances, IoT provides the ubiquitous connectivity among SCPs. Hence, e-services including remote monitoring, real-time simulation, and so on can be realized [8]. Cloud computing provides a SOA with different functional layers that enables ubiquitous access to a

shared pool of configurable system resources and higher-level services of end users in a “pay-per-use” business model [9]. Meanwhile, mobile edge computing, now officially named multi-access edge computing (MEC), leverages the computation power of large-scale smart devices [10]. Unlike cloud computing where data processing is conducted in the cloud or any central data warehouse, edge computing accelerates and improves its performance through data processing at the edge of a local network, near the source of the data [11]. CPS provides not only a hyper-connected environment, but also equipped with high-performance computational entities and physical hardware to achieve the smartness. The computational entities are connected with or embedded in the physical hardware so that they can monitor, control, operate and coordinate the hardware to achieve specific tasks [12]. Digital twins can be viewed as the virtual duplicate of physical products. The real-time situation of the technical entities can be reflected and the simulate operations can also be achieved in the virtual duplicate to trigger design innovations [13]. AI techniques, which leverage pervasive computing resources, can help to transform massive chaotic data into meaningful values to meet individual customer satisfaction [14].

3. Hybrid concerns for Smart PSS solution design

Smart PSS solution design is a complex process, which three general hybrid concerns are outlined below.

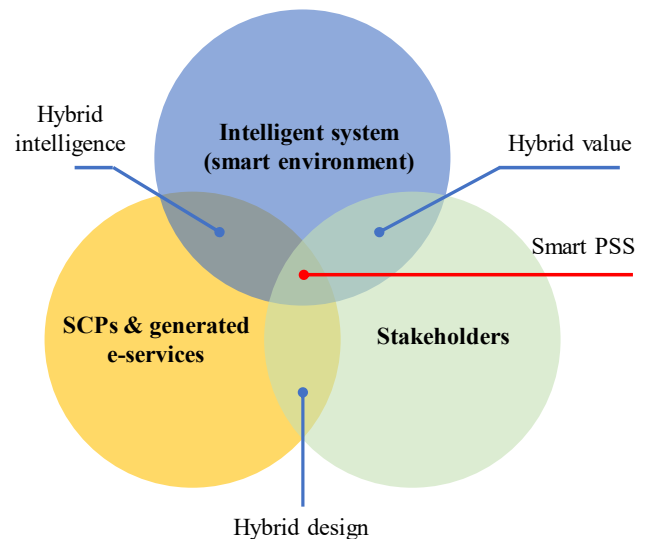


Fig. 1. Compositions of Smart PSS with hybrid concerns

3.1. Hybrid design

Unlike conventional product design approaches conducted in a mapping process between each domain (e.g. requirement domain to solution domain) sequentially or service design approaches in a more descriptive manner (e.g. service blueprint). In Smart PSS, it takes both design and redesign (inverse design) process into a hybrid consideration, i.e. *hybrid design*, by utilizing the advanced IT, where informatic-based (data-driven) approaches serve as the key to enable solution design success. Nevertheless, several challenges

remain not well-solved.

On one hand, in the early design phase, product-service family should be well-established in the IoT-enabled environment and adaptable to the dynamic changes of user requirements. On the other hand, design solutions are driven by the massive user-generated data and product-sensed data in the smart, connected environment and hence, should be capable to change/upgrade with context-awareness for redesign innovation/improvement during the usage stage [15].

3.2. Hybrid intelligence

As illustrated in our previous work [3], the major concern of conventional PSS lies in the efficient transmission of data/information (e.g. Internet) and it flourishes the prevailing e-commerce websites establishment. Crowdsourcing [16], as the practice of obtaining needed services or content by soliciting contributions from an online community, is a typical way for solution design with human intelligence solely. Meanwhile, IoT-enabled PSS is triggered by the ubiquitous connectivity of billions of mobile devices, vehicles, etc. where large-scale sensing devices are leveraged to obtain abundant data (e.g. crowd sensing) for design innovation purposes with machine intelligence. Nevertheless, the emerging Smart PSS cannot rely solely on the human intelligence or machine intelligence for reliable solution design [17]. Other types of data originated from service reports, maintenance records can also be utilized as a *hybrid intelligence*, to achieve better wisdom [1], which again is not reported.

3.3. Hybrid value

PSS originated from business value aspect, which emphasizes a servitized value proposition with less environment impact for a circular economy [18]. By nature, product lifecycle is extended with changeable/upgradeable intangible services other than its physical components, so that resources (e.g. manufacturing, logistics, etc.) can be saved with more sustainability. Similar to the concept of open innovation 2.0 [19], Smart PSS is a value co-creation business paradigm by utilizing cutting-edge IT. From industrial perspective, it can be seen as a sociotechnical eco-system, where users become an integral part of the innovation process (i.e. co-creation) and their experience, i.e. user experience, drives the innovation. Meanwhile, open solution design can be exploited as a principle that takes product open innovation, sustainability, intelligence and servitization into an overall consideration.

4. Prospective methodologies

To overcome those challenges, prospective methodologies are provided correspondingly from *design theory*, *intelligent system* and *value co-creation* aspects below, respectively.

4.1. Design theory aspect

Both system level design and product-service level design are considered as follows:

Product-service-context family design. In Smart PSS context, solution design (both design and redesign) should be

conducted with context-awareness. To achieve this, at system level, other than the conventional product-service family design, context (condition) information and the interrelationships among them should be considered as well. To be more specific, a solution can be modelled in a way $\langle P, C, S \rangle$ that “given a certain context (condition), what products and services should be triggered”, where product and service information can be decomposed into different modules/components, and conditions can be easily accessed through hardware sensors (product-sensed data) or social sensors (user-generated data) in an IoT-enabled manner [15]. By leveraging graph theories, each node represents the information of product, service or condition, while the edge represents the relationships in-between them.

Self-adaptable design. Product-service level design should follow adaptable design principles [20], while broaden its scope to both solution design considerations, by adding or replacing certain product-service modules through a pre-defined interface (e.g. API). Functional modelling is the key enabler for each design module to be independent of the others, so that the SCP and its e-services can be easily upgraded or changed based on individual user needs. Moreover, the solution design should be self-adaptable in the smart, connected environment with context-awareness [21] to allows certain degree of customer design freedom by embedding knowledge and rules about possible product or e-service differentiations into the SCP after manufactured, as *Smart PSS with built-in-flexibility*. The end solution can be adjusted continuously during its lifecycle usage in a data-driven manner.

4.2. Intelligent system aspect

As discussed before, intelligent system in this research refers to a broad scope of advanced IT to ensure high smartness and connectedness, including effective computing infrastructure, data collection and fusion techniques, and solution knowledge management aspects.

Cloud-edge computing infrastructure. Owing to the exponential increase of SCP introduced, it is predicted that there will be 50 billion smart devices connected to the Internet by 2020 with 507.5 ZB/year data generated by 2019 [25]. These big IoT data generated from distributed IoT devices cannot be properly handled by the remote cloud due to the large consumption of time, energy and bandwidth with high response time. Meanwhile, user privacy remains another critical issue [9] to be solved and in many cases, especially in manufacturing industry, stakeholders have strict regulations to protect their intellectual properties and business privacy in confidentiality. The emerging 5G networks and IoT provide better capability of data transmission and ubiquitous connectivity. Hyper-connected networks of SCPs can be created for information transmission. Meanwhile, advanced information processing techniques (AI) and increasing hardware computing capability (e.g. CPS) enable us to take full advantages of the heterogenous data/information sensed from various devices and generated from massive users to create many intelligent services [24]. In this context, end devices are not only data consumers from the cloud, but also data producers performing computing offloading, data storage,

caching and processing, as well as distribute request and delivery service from cloud to users [28]. By leveraging the advantages of both cloud computing and MEC, a hybrid IT infrastructure named cloud-edge computing [29] is proposed by considering the appropriate data processing and transmission between the edge and cloud.

Hybrid crowdsensing technique. Guo et al. [26] introduced a concept named mobile crowd sensing and computing (MCSC) by leveraging large-scale mobile devices and empowering massive users to share surrounding information or accomplish specific sensing tasks. Other than only collecting data from physical sensors, the participants, acting as the “social sensors”, have the ability to analyse data and transform into valuable knowledge with context-awareness [27]. Moreover, in the smart industrial environment, both the high reliability and performance of static sensing nodes/infrastructure (e.g. maintenance records derived) and the large-scale, cost-effective mobile devices with high mobility should be leveraged, as a hybrid community sensing approach, to generate new values.

Graph-based knowledge management. In the Smart PSS environment, heterogeneous data sources are converged consisting of unstructured, semi-structured and structured datasets. Hence, conventional relational databases are no longer suitable to handle such information as before. Graph database (Nosql ones) together with graph-based techniques should be deployed for the data storage, representation and querying. As mentioned before, three types of information, i.e. product, service, and context information are presented as the nodes, while their relationships are denoted as the weighted edges in-between them. Each data log (e.g. service record) is constructed by a condition set and decision set, respectively, where the former contains the condition nodes and the latter the ‘triggered’ service or product nodes, respectively. In order to derive solution design concept, graph theories such as community partition approach (e.g. Louvain method) or other clustering ones [1] can be exploited so as to automate the recommendation/prediction process with intelligence.

4.3. Value co-creation aspect

Specifically, in the Smart PSS context, value co-creation can be modelled as the mapping between *design solution space (DSS)* and the *user solution space (USS)* [28] before and after manufacture respectively, where:

- $DSS \rightarrow USS$: a manufacturer/service provider’s defined solution space drives a user-generated solution.
- $DSS \rightarrow DSS$: a manufacturer/service provider’s design solution urges self- or competitor-design innovation.
- $USS \rightarrow DSS$: a user-generated solution triggers the innovation of a manufacturer/service provider’s design solution.
- $USS \rightarrow USS$: one or more user-generated solutions contribute to other users’ innovations.

Both spaces are expanded jointly through design iteration, where two effective manners can be defined: platform-based and data-driven approaches. By establishing the virtual twin, SCPs and its e-service subscriptions can be reconfigured online during usage domain in the IoT-enabled platform as well by communicating with the physical product, and

adapted to meet individual needs remotely [29]. Meanwhile, SCPs own the powerful computation and communication capabilities, hence massive users are enabled to contribute their data/information through various channels (e.g. online communities, mobile social network, etc.), so that valuable knowledge can be extracted/generated readily for value generation purposes.

5. Industrial application scenarios

In order to make the proposed approaches more concrete, two potential industrial application scenarios are illustrated below with hybrid concerns.

5.1. Smart ordering system

Existing ordering systems undertake a configure, price and quote (CPQ) manner, where consumers normally select components/modules from the existing product-service family and got the quote to make a customized order. However, in the Smart PSS context, ordering system is becoming smarter and more personalized with high margins, following a ‘Configure $\rightarrow$ Instant Quote $\rightarrow$ Order $\rightarrow$ Track $\rightarrow$ Upgrade/Change $\rightarrow$ Postponed Manufacturing’ process, as shown in Fig. 2. It undertakes the customer-to-manufacturer (C2M) business model in a value co-creation manner. Owing to the product-service-condition family design and the real-time tracking in the CPS environment, smart ordering system allows postponed changes/upgrade (i.e. late differentiation) or even personalized add-ons with extra charges, before the component/module has been fabricated or assembled. Moreover, by adopting AI techniques, industrial supply-demand can be well-balanced to make accurate price/delivery time predictions to the consumers.

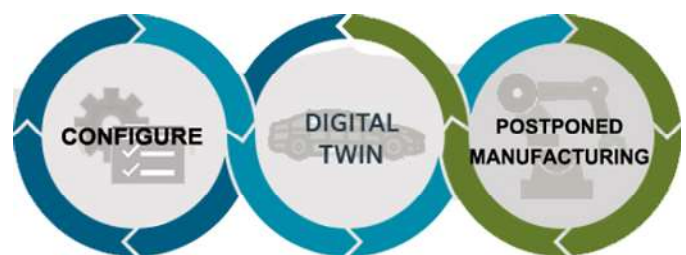


Fig. 2. Example of smart ordering system with late differentiation

5.2. Smart car maintenance and insurance case

Conventionally, car maintenance and insurance are conducted in either a face-to-face manner (offline) or online booking manner (online). Especially in the event of car accident, the car owner either call the police/insurance company to send people to the scene and claim duties or phone the repair shop to claim the money from the insurance company. Neither way is time consuming and costly. In the Smart PSS context, however, hybrid crowd sensing together with cloud-edge computing infrastructure can be leveraged in the future to intelligently benefit all parties dealing with the car maintenance and insurance services cost-effectively, as shown in Fig. 3. Taking car accident as an example, car

owners can take pictures of the scratches together with a claim report through the specific mobile APP installed in the smart phone. Those data stream will be sent to the MEC layer with edge nodes to notify local repair shops nearby based on GPS location information, and compute the cost instantly based on the image processing and text mining techniques. Meanwhile, all the accident claim datasets will be uploaded to the insurance company cloud remotely through the software development networks (SDN) for big data analytics to achieve better insurance policy plan in the near future. Those new service policies will be pushed to the end mobile devices from the cloud as service applications (normally an APP) with add-on values for users' subscription.

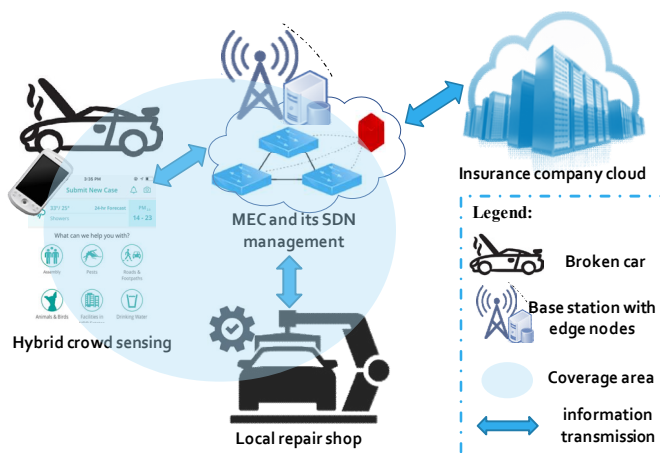


Fig. 3. Example of hybrid crowd sensing with cloud-edge computing infrastructure for smart car maintenance and insurance.

6. Conclusions

Smart PSS is an emerging topic by embracing the advanced IT innovations. Industrial companies in such context can largely benefit from this IT-driven value co-creation business strategy to deliver 'evergreen' design solutions with extended product-service lifecycle and better customer satisfaction. This research, as an explorative study, outlines three generic concerns to distinguish Smart PSS from other PSS paradigms in the solution design stage, i.e. hybrid design, hybrid intelligence, and hybrid value. To overcome these challenges, prospective methodologies are introduced from design theory aspect (i.e. product-service-condition family design, and adaptable and self-adaptable design), intelligent system aspect (i.e. cloud-edge computing infrastructure, hybrid crowdsensing, and graph-based knowledge management), and value co-creation aspect (i.e. solution mapping), respectively. To present its feasibility, two promising industrial applications are further illustrated in the Smart PSS context. Despite these contributions, due to the page limit, more in-depth works should be further conducted in each of the above-mentioned research areas or from other perspectives to achieve the solution design success, and it is hoped this research can offer insightful guidance to industrial companies in their solution design process with high revenue.

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